CLAIMS

What is claimed is:

1. A method for representing an analog signal in a digital system, comprising the steps of:
   - sampling the analog signal at a plurality of discrete time intervals to produce a sequence of analog samples; and
   - for each analog sample in the sequence, directly mapping a characteristic of the sample to a corresponding discrete state in a set of more than two available states of a parallel data interface, thereby creating a sequence of high-cardinality data words.

2. The method of claim 1, wherein the parallel data interface is an optical interface, and the discrete states are different wavelengths or colors of light.

3. The method of claim 1, wherein the characteristic of the sample is its amplitude.

4. The method of claim 1, wherein the sequence of high-cardinality data words is transmitted as a single frame of data over the parallel data interface.

5. The method of claim 1, wherein the analog signal is an audio signal.

6. A system for representing an analog signal, comprising:
   - a sampling module configured to sample an analog signal at a plurality of discrete time intervals to produce a sequence of analog samples;
   - a direct encoding module configured to receive the sequence of analog samples and, for each sample, map a characteristic of the sample to a corresponding discrete state in a high-cardinality state space of a parallel data interface; and
   - a parallel data interface configured to transmit the resulting sequence of high-cardinality states.

7. The system of claim 6, wherein the parallel data interface is an optical interface, and the discrete states are different wavelengths of light.